

INTRALUMINAL DISTANCE/GEOMETRY DEVICE USING ULTRASOUND FOR ENDPOINT DETECTION, RANGE MEASUREMENT, AND INTERNAL FEATURE LOCALIZATION (NON-LIMITING)

A. Technical Field

This section relates to an insertable intraluminal device (Node 4) configured to measure internal distance and/or geometry using ultrasound sensing as an endpoint detection method, including time-of-flight and/or echo-amplitude methods, and to optionally characterize compliance/contact under controlled radial stimulus.

B. Summary (Non-Limiting)

In various embodiments, Node 4 is an intraluminal insertable device (intravaginal and/or anal) that measures internal distance to an internal endpoint and/or internal geometric features. While optical time-of-flight (IR ToF) may be used in certain embodiments, other embodiments use ultrasound as an endpoint sensing modality to improve robustness across diverse tissue conditions, moisture states, and internal surface properties. Node 4 may include: 1. an ultrasound transducer (single-element or array); 2. a drive and receive front-end configured to transmit ultrasound pulses and detect echoes; 3. signal processing configured to estimate endpoint distance using time-of-flight, echo envelope detection, correlation, and/or thresholding; and 4. optional controlled radial stimulus elements (e.g., inflatable cuff/microbladder ring, constraint band, mechanical cam, or micro-pump actuation) with pressure sensing and circumferential contact sensing to characterize compliance and contact.

C. Ultrasound Endpoint Measurement Principles (Non-Limiting)

In various embodiments, Node 4 determines a distance to an internal endpoint or anatomical boundary by transmitting one or more ultrasound pulses and analyzing a reflected signal. Non-limiting distance estimation methods include: 1. pulse-echo time-of-flight distance estimation based on the round-trip travel time of echoes; 2. echo envelope detection to identify a dominant reflecting boundary; 3. multi-pulse averaging and confidence scoring to reduce noise; 4. matched filtering or correlation to improve timing accuracy; and 5. signal quality gating to suppress unreliable windows. In certain embodiments, the system computes a distance metric as an index or range (rather than raw imagery), and reports confidence indicators.

D. Ultrasound Transducer and Placement (Non-Limiting)

In various embodiments, the ultrasound transducer is positioned near a distal tip region of the device and oriented along a longitudinal axis to interrogate an internal endpoint direction. Non-limiting configurations include: 1. single-element piezoelectric transducer configured for pulse-echo operation; 2. multiple transducers to provide redundancy and angle diversity; 3. a small array for directional sensitivity and improved endpoint discrimination; and 4. transducer mounting behind a thin acoustically transmissive window. In certain embodiments, the transducer is sealed within a soft/biocompatible tip and is configured to couple acoustically through a compliant layer.

E. Coupling Medium and Acoustic Interface (Non-Limiting)

In various embodiments, ultrasound operation is improved by an acoustic coupling interface. Non-limiting coupling approaches include: 1. a hydrophilic surface or coupling-friendly tip material; 2. user-applied coupling medium; 3. a built-in micro-reservoir or surface film designed to support consistent coupling; and 4. coupling plausibility checks using echo amplitude, stability, and/or temperature context. In certain embodiments, the system outputs an acoustic coupling quality indicator and suppresses distance outputs if coupling is insufficient.

F. Signal Processing and Endpoint Selection (Non-Limiting)

In various embodiments, the computing system identifies an endpoint echo and computes distance using: 1. envelope detection and thresholding; 2. selecting the first strong reflector beyond a minimum range; 3. rejecting spurious reflections using continuity constraints across sequential pulses; 4. IMU-based motion gating to reduce artifact; and 5. confidence scoring based on echo stability, amplitude, and temporal coherence. In certain embodiments, the system uses multiple pulses at different amplitudes or repetition rates to improve robustness and detect signal saturation.

G. Integration with Compliance/Contact Characterization (Optional, Non-Limiting)

In optional embodiments, Node 4 includes controlled radial stimulus (e.g., cuff/microbladder inflation) and pressure sensing. The system may: 1. apply a controlled inflation profile; 2. measure stimulus pressure using a MEMS pressure transducer; 3. measure circumferential contact using an FSR/capacitive pressure ring; and 4. compute compliance/contact indices while simultaneously tracking ultrasound endpoint distance stability. This enables a combined profile including distance-to-endpoint metrics and compliance/contact metrics.

H. Privacy-Preserving Outputs (Non-Limiting)

In various embodiments, ultrasound-derived outputs are presented as abstract metrics (indices, ranges, bars, curves, and confidence indicators) and do not require generating or displaying anatomical images. Raw echo waveforms may be processed on-device and discarded after feature extraction, unless explicitly enabled.

I. Variations

Any embodiment described herein may use different ultrasound frequencies, pulse patterns, transducer configurations, coupling interfaces, and processing methods while maintaining the functional intent of ultrasound-based endpoint detection and distance estimation within an intraluminal device.